

Hunter Hawkins

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CAREER OBJECTIVE

Innovative PhD candidate in Computer Science specializing in explainable artificial intelligence and machine learning applications for industrial automation and infrastructure. Leveraging an industry focused foundation in automation to bridge the gap between research and real-world implementation, driving smarter, more resilient, and efficient systems across industrial environments.

INDUSTRY EXPERIENCE

Wapiti Consulting

May 2021- Present

Software Developer/ Control System Integrator

Coeur d'Alene, ID

- Developed custom real time web based SCADA systems using the Python Django web framework.
- Manage all IT related infrastructure for Wapiti Consulting, 3R Valve, Panelview RV Park and others
- Integrated industrial PCs with Allen-Bradley CompactLogix and Schneider Electric Modicon PLCs.
- Assisted in taking Panelview RV Park from a \$1 million dollar investment to a \$4 million dollar evaluation with custom software solutions including autonomous check-in, smart keypads, power and water monitoring.
- Establish cloud based MQTT communications between remote sites using industrial PCs communicating with PLCs
- Wrote data collection software for Aquifer Pumped Hydro pilot project in Kern County, California.
- PLC, HMI, and VFD software development for Allen-Bradley, Schneider Electric, Red Lion, and other vendors
- Manage two other software developers hired from UI CDA and co-manage technician assembly team.
- Maintain a customer service focused attitude towards operators and municipalities.
- Design, build, and troubleshoot UL 508A industrial control panels | UL508A certified

NASA

May 2020 – May 2021

Software Engineer Intern (NESSP)

Remote

- Led the initial design/development of a robot controlled lunar lava tube sealant process via a hoberman sphere.
- Recruit and manage a team from North Idaho College and assist with technical features of their rover.

University of Idaho Mathematics/ Computer Science

September 2019 – May 2021

CS/ Math Lab Teacher

Moscon, ID

- Assisted fellow students in basic to complex problems in computer science and mathematics.

Kootenai Girls Softball Association

May 2012 – July 2019

Umpire

Coeur d'Alene, ID

Tomato Street CDA

June 2016 – August 2019

Busser / Dishwasher

Coeur d'Alene, ID

ACADEMIC AND PERSONAL PROJECT EXPERIENCES

Amalgamated Sugar Explainable AI PhD Project

August 2023 -

Present

P.h.D Researcher

Coeur d'Alene, ID

- Gathered data from the dataPARC historian and formatted it with the python libraries of numpy and polars
- Trained 4 models on the data including a decision tree model, custom time offset decision tree model, a long short term memory (LSTM) model, and a hybrid model combining a decision tree and LSTM
- Performed custom time scoring on trained models with best models predicting over 75% of issues in advance
- Built a custom django dashboard that would display issues to the operators including the ability to dynamically change between models.
- Integration with a OPC UA server to pull real time data

- Worked closely with Amalgamated Sugar process engineers to ensure success of project

NIATT/ PacTrans/ U.S. DOT MS Project

August 2021 – May 2023

M.S. Researcher

Coeur d'Alene, ID

- Developed a survey paper aimed at helping small municipalities implement smart city projects with minimal expertise (IoT Platforms for Smart City Implementation in Rural and Urban Communities: A comprehensive Review)
- Worked with multidisciplinary team including a civil engineering and cybersecurity advisor
- Publication by U.S D.O.T and Pactrans and conference presentation

Personal Website

August 2021 – Present

Personal Project

Coeur d'Alene, ID

- Personal project built with Django web development framework
- Includes custom weather application using the OpenWeatherMap api

EDUCATION

University of Idaho

May 2019 - Present

P.h.D Computer Science

Coeur d'Alene, ID

- 3.95/4.0 GPA, Emphasis in machine learning and artificial intelligence
- Graduate Certificate in Robotics and Automation
- Project lead on Amalgamated Sugar Steam Dryer AI Project
- Design, build and maintain all lab equipment for PLC laboratory
- Design and teach cross disciplinary Advanced PLC class from scratch, assist in teaching PLCs 2 semesters

M.S Computer Science

Coeur d'Alene, ID

- 4.0/4.0 GPA, Emphasis in critical infrastructure, smart cities, IoT, and embedded systems
- Masters Paper published by PacTrans/ USDOT

B.S Computer Science and B.S Mathematics

Moscon, ID

- 3.5/4.0 cumulative GPA, Dean's List, Pi Mu Epsilon
- Event coordinator for the automotive enthusiasts club and member of math club

North Idaho College

August 2016 – May 2019

AS, Computer Science

Coeur d'Alene, ID

- Phi Theta Kappa, Dean's List
- Computer Science and Technology Club Member 2017-2019.

SKILLS, INTERESTS and CERTIFICATIONS

- **Skills:** AI/ ML, Python Programming, Networking, PLC/ HMI/ VFD Programming, Django, Industrial Controls, Systemd, Electrical Wiring, MQTT, Systems Integration, SCADA
- **Interests:** Automation, Software Development, Investing, MMA, Jiu Jitsu, Cars/ Motorcycles
- **UL508A MTR Certification:** Underwriters Laboratories standard for safe construction of industrial control panels. Transformers, branch circuits, SCCR, Motor FLA, etc.
- **University of Idaho Robotics and Automation Certification:** Took courses on Robotics, SCADA, PLC programming and Real Time Operating Systems.